PEER REVIEW HISTORY

BMJ Open publishes all reviews undertaken for accepted manuscripts. Reviewers are asked to complete a checklist review form (**http://bmjopen.bmj.com/site/about/resources/checklist.pdf**) and are provided with free text boxes to elaborate on their assessment. These free text comments are reproduced below.

ARTICLE DETAILS

TITLE (PROVISIONAL)	Assessing the effectiveness of workplace accommodations in facilitating return-to-work after traumatic brain injury – a systematic review protocol.
AUTHORS	Shafi, Reema; Colantonio, Angela

VERSION 1 – REVIEW

REVIEWER	Ben Bloom Barts Health NHS Trust Queen Mary University of London UK
REVIEW RETURNED	23-Jul-2020

GENERAL COMMENTS	This is an important topic and clearly fills a gap in knowledge. The authors have scrupulously followed the PRISMA-P guidelines. One main barrier to this review generating a single metric, which the authors clearly anticipate, is that workplace accommodations are likely to be numerous and heterogenous, and thus only reportable but not able to be pooled. That they plan to include qualitative research is I believe one strategy to get around this problem. The other barrier to success may be that there is insufficient data published on this, however that finding in itself would be critically important in improving the working lives of patients that have had a TBI. The stages of study identification are written in the narrative and when the review is complete should be graphically represented in a figure 1. I suggest the figure 1 (without results) be drafted so that readers of the protocol can see at a glance how studies were included. The quality assessment section of the protocol is relatively brief. The authors plan to use CASP checklists because there are different checklists for various types of study. There is no consensus on the 'best' tool to use when reporting bias, and it's generally agreed that as long as bias is assessed, the specific tool used is of less consequence, and therefore CASP is as good as any. I suggest describing how inter-study bias will be reported - narrative/table/graphically? Otherwise pleased to accept this manuscript for publication.

REVIEWER	Mirinda Gormley NA, United States of America
REVIEW RETURNED	04-Aug-2020

GENERAL COMMENTS	This is a really well-written protocol, and I would very much look forward to reading the full review. I agree that this is a very

	important topic for discussion, and that WAs are a very important tool to help re-integrate survivors of TBI back into the workplace. I believe that the information that could be provided by this systematic review would be very valuable to the development of accommodations that allow individuals to return to productivity following TBI. Below are several additional comments that I believe that the authors may want to consider as they continue their review. Page 2, Limitations: I think there are quite a few more limitations that the authors will have to consider while conducting this review. Like much of the treatment for TBIs, WAs are likely individualized to the needs of each specific patient, and vary depending on the occupation where they are being applied. Thus, in addition to the heterogeneity of the TBI population, the heterogeneity in the types of WAs, and the studies on WAs will also limit the author's ability to make precise interpretations. Additional limitations (country of study, variation in disability provisions) are mentioned in more detail below. Page 6 Line 217: You state that all types of employment will be considered (full-time, part-time, casual, contract, self-employed, volunteer, etc.). Some previous studies have described "education" as a sort of employment; will you consider including individuals that are returning to full-time education or who are receiving accommodations to return to education in this review? If not, perhaps that should be mentioned as a reason for exclusion. Page 6 Lines 220-221: If the data are sufficient, might you also consider stratifying by the country of origin for the study participants/data? Disability benefits vary widely from country to country, so it would be reasonable to expect variation in disability accommodations between countries as well. For example, you mention the CHRC on page 7, pointing out that (in Canada) the employer has a duty to take reasonable steps to accommodate the position of an employee unless the business would suffer 'undue hardship'…to my knowledge there is no law equivalent to this in the United States. This variation is often listed in other reviews as a limitation, as it is difficult to account for whether someone decided to receive disability benefits or return to work out of financial necessity. If you choose not to stratify for country of origin, how would you intend to address this issue? Page 6 Lines 227-234: While publication year is important to identify the studies in the review, are you also intending to include the years the study was performed? E.g., it would be helpful for the reader to know that data was analyzed from the 2012-2018 TBIMS for a study published in 2020. It is possible you intend to include this information in the study setting, I only ask because it isn't specifically listed. Page 6 Lines 227-234: If data was sufficient, might it also be possible to indicate whether the individuals returned to pre-injury employment vs. individuals who are returning to work in general? Understanding the difference between individuals who are able to return to pre-injury level of work and those who are able to find employment in general following TBI could reveal the occupation types where accommodations are less prevalent, and could uncover areas for additional research. Further, in your discussion section (page 8 lines 312-314), you point out that one of the

	benefits for WAs is the “retention of previously trained employees”, however this statement cannot be true of all WAs, as not all individuals are able to return to their pre-injury employment. I think it is very important given this rationale for the importance of your study, that you consider where/how you can differentiate between the studies where individuals are returning to pre-injury level of work and those returning to any level of productivity. Page 6 Lines 234-236: This sentence confuses me. Are you stating that a secondary table will be created that reports these secondary outcomes, or are you saying these secondary outcomes will be included in a table with conflicts of interest and ‘risk of bias’ assessments? The risk of bias assessments should be included in a table on its own, independent of secondary outcomes. While a risk of bias score that comes from the CASP checklist could be inserted into a table that also contains conflicts of interest, it doesn’t make sense to me why secondary outcomes should be reported with this information. I would suggest just making a smaller secondary table with the secondary outcomes, and would also suggest that the overall risk of bias score somehow be incorporated into the large, main table. Page 8: Lines 301-303: If your review intends to distinguish the difference between WAs and ‘natural supports’, then these ‘natural supports’ should be defined above in the methods section, and you should indicate how they will be extracted and where they will be discussed in your review. Page 8 Line 315-326: I think this would be a good place to acknowledge that the final statement of this paragraph does not apply to all countries. While the information in this study would be very relevant to the HCPs, clinicians, and employers in countries where they are required to apply accommodations if possible (Canada), or claim “fitness to work” (in the UK), how could this information inform the HCPs, clinicians, and employers in the United States, where many states are right-to-work states and individuals may be provided with little to no disability benefits? The prevalence of TBI is very high in the U.S., and I predict that many of the studies included in this review will come from the U.S., so I believe it is important to address how these findings would be useful to stakeholders in countries with legislation that is less accommodating to individuals with disabilities. Page 8-page 9, lines 335-341: I don't believe this small section is an appropriate rationale for this protocol. The age-criteria for your study was 18-65, which you stated was “working age”…how can you justify the usability of the findings of this review by stating that Effective WAs ”….create opportunities for the aging population by enabling their productivity post-retirement” if you aren’t including studies of individuals of retirement age? I would recommend omitting these lines. Page 9 lines 344-346: Please reconsider the comma placements in this final sentence.

REVIEWER	Daniele Evelin Alves Oslo Metropolitan University, Work Research Institute, Oslo, Norway
REVIEW RETURNED	07-Aug-2020

GENERAL COMMENTS	Review report: Assessing the effectiveness of workplace accommodations in facilitating return-to-work after brain injury – a systematic review protocol. Recommendation: Major revision Reviewer's summary of the article: The protocol is well-written and addresses an important research gap in the literature: The effectiveness of different specific work accommodations on return-to-work following acquired brain injury. However interesting and important, I have some important concerns related to the operationalization of terms in a multi-disciplined research field, as well as the ambitious scope of the future review. Firstly, the role of work accommodation in facilitation return-to-work among employees with acquired brain injury has been studied in various disciplines such as medicine, psychology, social work, rehabilitation, etc… Disciplines use different words to study work accommodation and return-to-work. It would be helpful to readers if the authors could provide a more extensive list of subject headings and keywords, and explain how exactly they were combined (for instance: how was the PICO framework modified, and how were words combined? Did you include "work retention" OR "return-to-work" OR "work tenure")? This could be done by including an appendix, or explaining the search words and strategy more clearly on line 161. As one of the authors of a recent systematic review on work and workplace characteristics related to work retention/return-to-work among employees with brain injuries, I was initially surprised by the wide variation of terms necessary to capture our research question across disciplines when conducting our data search. Our review (Alves et al., 2020) was not cited in the present protocol, and I strongly recommend the authors to read it as well as its reference list when revising the protocol. My intention is not to misuse my role as a reviewer in order to be cited, but to make sure that the protocol is up-to-date and complete. As far as I can see, Alves et al (2000, se full reference below) includes updated research that should be included in the protocol, as it includes work accommodations and brain injury (even though its scope is wider in terms of exposure and target group). The background section could be more precise in terms of clarifying literature on work accommodations including groups other than employees with TBI (e.g., lines 112-114, perhaps include "among employees with other injuries than TBI" after the sentence to emphasize that Kristman and colleagues did not study TBI). Consider including "acquired" before "brain injury" in the title or early in the abstract to be more precise about the target group. Consider clarifying what is meant by "both stakeholders" on line 108 (eg., "employers and employees"). Why do you not include only employees in paid/competitive employment? Are work accommodations including volunteer work comparable to those including a salary? (lines 217-218). Secondly, the protocol describes ambitious goals (lines 147-151). Synthesizing both quantitative and qualitative evidence in the field is no easy task. This leaves me with questions as regards data search (Will the authors be overwhelmed by the number of studies to review as "work accommodations" can be quite broadly defined across fields? How will you deal with all search hits in a systematic way?). Please consider using the software Covidence. I did not understand how you will synthesize findings from high-quality and low-quality studies across quantitative and qualitative

	methods: will findings weigh differently when you synthesize them (e.g., will findings from a high-quality case study with one individual be more emphasized than those from a low-quality longitudinal study?) And how will you use statistics in quantitative studies, and what do you mean by “significance is multi-fold”, line 42? If, after considering these questions, the authors become worried about the complexity and time needed to complete a mixed-method review, please consider dividing the review into two: one quantitative and one qualitative. Finally, please reformulate the discussion to reflect what we do not know yet about the effectiveness of work accomodations (eg., lines 339-341: from “lessens the economic burden”, “fosters increased productivity” to “may lessen” and “may foster”). Work accommodations may not always increase workplace productivity as some employees need substantive follow-up from co-workers and longer time to perform tasks after TBI. Also, work accommodations in volunteer work (which is included in the present protocol) are not expected to lessen the economic burden of employees after TBI. All in all, I believe that this protocol and its future review may become an important contribution to our research community. Being more open about the limitations related to the enormous scope of a mixed-method review (and clear about how they plan to handle potential problems of time and resources- what is plan B if plan A goes wrong?), and be more transparent about search words, combinations, and quality appraisal and synthesis. Good luck with this important work. Full reference of recent review: Alves DE, Nilsen W, Fure SCR, et al. What characterises work and workplaces that retain their employees following acquired brain injury? Systematic review. Occup Environ Med 2020;77(2):122-30.

VERSION 1 – AUTHOR RESPONSE

Reviewer #1 - Ben Bloom

1. One main barrier to this review generating a single metric, which the authors clearly anticipate, is that workplace accommodations are likely to be numerous and heterogenous, and thus only reportable but not able to be pooled. That they plan to include qualitative research is I believe one strategy to get around this problem.
2. The other barrier to success may be that there is insufficient data published on this, however that finding in itself would be critically important in improving the working lives of patients that have had a TBI.

Our response: We recognize the heterogeneous nature of workplace accommodations given the heterogeneous nature of the injury (TBI) as well as personal and vocational factors (organizational structure, work industry, employment type etc.). It has never been our objective to outline a single metric for workplace accommodations as that would undermine the richness of the existing data. We intend to separately assess the quantitative and qualitative data regarding workplace accommodations. We are also open to present and analyze the data using a mixed methods analysis approach, if necessary. I have revised the narrative on page 4, line 152. We agree that the recognition regarding the paucity of data, if that is the case, will identify a key knowledge gap.

3. The stages of study identification are written in the narrative and when the review is complete should be graphically represented in a figure 1. I suggest the figure 1 (without results) be drafted so

that readers of the protocol can see at a glance how studies were included.

Our response: We acknowledged and thank you for the suggestion and will be considering it as we complete our review.
4. The quality assessment section of the protocol is relatively brief. The authors plan to use CASP checklists because there are different checklists for various types of study. There is no consensus on the 'best' tool to use when reporting bias, and it's generally agreed that as long as bias is assessed, the specific tool used is of less consequence, and therefore CASP is as good as any. I suggest describing how inter-study bias will be reported - narrative/table/graphically?

Our response: Inter study bias will be reported separately in a table and may be discussed in the narrative.

Reviewer #2 - Mirinda Gormley

1. Page 2, Limitations: I think there are quite a few more limitations that the authors will have to consider while conducting this review. Like much of the treatment for TBIs, WAs are likely individualized to the needs of each specific patient, and vary depending on the occupation where they are being applied. Thus, in addition to the heterogeneity of the TBI population, the heterogeneity in the types of WAs, and the studies on WAs will also limit the author's ability to make precise interpretations. Additional limitations (country of study, variation in disability provisions) are mentioned in more detail below.

Our response: On page 2 (lines 54-58), we have modified the statements to included additional limitation pertaining to the issues emphasized by the reviewers. They read as follows:
• Heterogeneity of the traumatic brain injury population, uniqueness of the employer/occupational variables and individualized preferences and specific needs of the worker may pose a challenge for more precise interpretations.
• Environmental variables such as country of origin and, more specifically, variations in disability provisions, will need to be considered when interpreting the findings.

2. Page 6 Line 217: You state that all types of employment will be considered (full-time, part-time, casual, contract, self-employed, volunteer, etc.). Some previous studies have described "education" as a sort of employment; will you consider including individuals that are returning to full-time education or who are receiving accommodations to return to education in this review? If not, perhaps that should be mentioned as a reason for exclusion.

Our response: Yes, we plan on including education in this category as it is a marker of productivity in individuals. We have revised the line of interest as follows:
With respect to vocational status, all types of employment (full-time, part-time, casual, contract, self-employed, students, volunteer etc.) and occupational categories will be considered.
3. Page 6 Lines 220-221: If the data are sufficient, might you also consider stratifying by the country of origin for the study participants/data? Disability benefits vary widely from country to country, so it would be reasonable to expect variation in disability accommodations between countries as well. For example, you mention the CHRC on page 7, pointing out that (in Canada) the employer has a duty to take reasonable steps to accommodate the position of an employee unless the business would suffer 'undue hardship'…to my knowledge there is no law equivalent to this in the United States. This variation is often listed in other reviews as a limitation, as it is difficult to account for whether someone decided to receive disability benefits or return to work out of financial necessity. If you choose not to stratify for country of origin, how would you intend to address this issue?
4. Page 6 Lines 227-234: While publication year is important to identify the studies in the review, are you also intending to include the years the study was performed? E.g., it would be helpful for the

reader to know that data was analyzed from the 2012-2018 TBIMS for a study published in 2020. It is possible you intend to include this information in the study setting, I only ask because it isn't specifically listed.

Our response: Yes, these variabilities are of interest. Our current data extraction sheet is designed to collected information for the country of origin as well as the time the study was conducted.
5. Page 6 Lines 227-234: If data was sufficient, might it also be possible to indicate whether the individuals returned to pre-injury employment vs. individuals who are returning to work in general? Understanding the difference between individuals who are able to return to pre-injury level of work and those who are able to find employment in general following TBI could reveal the occupation types where accommodations are less prevalent, and could uncover areas for additional research. Further, in your discussion section (page 8 lines 312-314), you point out that one of the benefits for WAs is the "retention of previously trained employees", however this statement cannot be true of all WAs, as not all individuals are able to return to their pre-injury employment. I think it is very important given this rationale for the importance of your study, that you consider where/how you can differentiate between the studies where individuals are returning to pre-injury level of work and those returning to any level of productivity.

Our response: Yes, absolutely! We recognize that this distinction is an important one. Where possible we would like to report if this was a return to work at the pre-injury job, return to work at the pre-injury job with modification and/or accommodation, a return to work at a new job (with without added training and/or accommodations), the first job for this worker (e.g., student or unemployed at time of injury) etc. Our data extraction sheet is designed to be sensitive to this information.
6. Page 6 Lines 234-236: This sentence confuses me. Are you stating that a secondary table will be created that reports these secondary outcomes, or are you saying these secondary outcomes will be included in a table with conflicts of interest and 'risk of bias' assessments? The risk of bias assessments should be included in a table on its own, independent of secondary outcomes. While a risk of bias score that comes from the CASP checklist could be inserted into a table that also contains conflicts of interest, it doesn't make sense to me why secondary outcomes should be reported with this information. I would suggest just making a smaller secondary table with the secondary outcomes, and would also suggest that the overall risk of bias score somehow be incorporated into the large, main table.

Our response: We have reworded this sentence to avoid the confusion. Yes, the secondary outcomes will be presented under a separate table.
7. Page 8: Lines 301-303: If your review intends to distinguish the difference between WAs and 'natural supports', then these 'natural supports' should be defined above in the methods section, and you should indicate how they will be extracted and where they will be discussed in your review.

Our response: Within the manuscript, we have clarified regarding the limited scope of this review to include workplace accommodations only; hence, we have clarified it as such under discussion, page 8, lines 306-307.

8. Page 8 Line 315-326: I think this would be a good place to acknowledge that the final statement of this paragraph does not apply to all countries. While the information in this study would be very relevant to the HCPs, clinicians, and employers in countries where they are required to apply accommodations if possible (Canada), or claim "fitness to work" (in the UK), how could this information inform the HCPs, clinicians, and employers in the United States, where many states are right-to-work states and individuals may be provided with little to no disability benefits? The prevalence of TBI is very high in the U.S., and I predict that many of the studies included in this review will come from the U.S., so I believe it is important to address how these findings would be useful to stakeholders in countries with legislation that is less accommodating to individuals with

disabilities.

Our response: As requested, we have made this revision.
9. Page 8-page 9, lines 335-341: I don't believe this small section is an appropriate rationale for this protocol. The age-criteria for your study was 18-65, which you stated was “working age”…how can you justify the usability of the findings of this review by stating that Effective WAs ”….create opportunities for the aging population by enabling their productivity post-retirement” if you aren’t including studies of individuals of retirement age? I would recommend omitting these lines.

Our response: Thanks, we have omitted these lines.
10. Page 9 lines 344-346: Please reconsider the comma placements in this final sentence.

Our response: We have omitted the commas.

Reviewer #3 - Daniele Evelin Alves

Recommendation: Major revision
1. It would be helpful to readers if the authors could provide a more extensive list of subject headings and keywords, and explain how exactly they were combined (for instance: how was the PICO framework modified, and how were words combined? Did you include “work retention” OR “return-to-work” OR “work tenure”)? This could be done by including an appendix, or explaining the search words and strategy more clearly on line 161.

Our response: In our submission, on June 18, 2020, we included the complete search strategy which included the search terms, subject headings and keywords used as well as how they were combines. Please feel free to share with the reviewers, if not already.
2. As one of the authors of a recent systematic review on work and workplace characteristics related to work retention/return-to-work among employees with brain injuries, I was initially surprised by the wide variation of terms necessary to capture our research question across disciplines when conducting our data search. Our review (Alves et al., 2020) was not cited in the present protocol, and I strongly recommend the authors to read it as well as its reference list when revising the protocol. My intention is not to misuse my role as a reviewer in order to be cited, but to make sure that the protocol is up-to-date and complete. As far as I can see, Alves et al (2000, se full reference below) includes updated research that should be included in the protocol, as it includes work accommodations and brain injury (even though its scope is wider in terms of exposure and target group).

Our response: Thank you for bringing this recent work to our attention. With due respect, the paper cited is not focussed on workplace accommodation and as stated has a wider scope and target group. We recognize the paper as an important contribution to the field and intend to cite it in context when discussing our systematic review findings.
3. The background section could be more precise in terms of clarifying literature on work accommodations including groups other than employees with TBI (e.g., lines 112-114, perhaps include “among employees with other injuries than TBI” after the sentence to emphasize that Kristman and colleagues did not study TBI).

Our response: We have revised this statement.
4. Consider including “acquired” before “brain injury” in the title or early in the abstract to be more precise about the target group.

Our response: We have now used the word “traumatic” in front of “brain injury” in the title to be consistent with the abstract and the manuscript.
5. Consider clarifying what is meant by “both stakeholders” on line 108 (eg., “employers and

employees").

Our response: We have made this revision.
6. Why do you not include only employees in paid/competitive employment? Are work accommodations including volunteer work comparable to those including a salary? (lines 217-218).

Our response: The intent is to be inclusive and to comprehensively capture how workplace accommodations can promote re-entry into the workforce, irrespective of remunerative outcomes.
7. Secondly, the protocol describes ambitious goals (lines 147-151). Synthesizing both quantitative and qualitative evidence in the field is no easy task. This leaves me with questions as regards data search (Will the authors be overwhelmed by the number of studies to review as "work accommodations" can be quite broadly defined across fields? How will you deal with all search hits in a systematic way?). Please consider using the software Covidence.

Our response: Workplace accommodations will be only be screened during level II (full text) screening. The team will utilize a customized-designed semi-automated excel worksheet for the purpose, however, Covidence will be considered if the team members mutually agree upon it.

8. I did not understand how you will synthesize findings from high-quality and low-quality studies across quantitative and qualitative methods: will findings weigh differently when you synthesize them (e.g., will findings from a high-quality case study with one individual be more emphasized than those from a low-quality longitudinal study?)
9. And how will you use statistics in quantitative studies, and what do you mean by "significance is multi-fold", line 42? If, after considering these questions, the authors become worried about the complexity and time needed to complete a mixed-method review, please consider dividing the review into two: one quantitative and one qualitative.

Our response: We will divide the review into two manuscripts, one consolidating the quantitative and another the qualitative studies.
10. Please reformulate the discussion to reflect what we do not know yet about the effectiveness of work accomodations (eg., lines 339-341: from "lessens the economic burden", "fosters increased productivity" to "may lessen" and "may foster"). Work accommodations may not always increase workplace productivity as some employees need substantive follow-up from co-workers and longer time to perform tasks after TBI. Also, work accommodations in volunteer work (which is included in the present protocol) are not expected to lessen the economic burden of employees after TBI.

Our response: We have made this revision.

Lastly, we thank you for the thorough review and for considering our manuscript for publication. We look forward to your response.

VERSION 2 – REVIEW

REVIEWER	Mirinda Gormley Clemson University, United States
REVIEW RETURNED	17-Nov-2020

GENERAL COMMENTS	Reviewer Comments: First of all, I want to commend the authors for writing this review. Overall, I think that the review is a really good idea – a comprehensive review of workplace accommodations for individuals with TBI would be invaluable to engaging legislative

	efforts around this issue, as well as those considering starting their own program. While I believe that this protocol is worth publishing, there are major revisions that still need to be addressed before this manuscript is ready for publication. I double checked several of my first comments against the marked version of the draft and was surprised how few changes were incorporated into the document. I asked several questions about the data extraction portion, specifically adding more detail to whether country of origin would be captured, and years of data collection. You responded that the variables would be included in the extraction sheet, yet without access to that extraction sheet my question remains. It would be helpful if the data extraction section included these details. For example, what will be captured for study setting? Is this going to be a description of the population, the years the study was conducted, the university conducting the study? I would suggest revising this to include those details "…study setting (e.g., source of population, years for data analyzed, country of study, etc.)" Similarly, I asked the how you would distinguish between "return to work" and "return to pre-injury work", as I believe that would provide important context for your discussion. The response was that the data extraction sheet was designed to be sensitive to this information, however there were no updates made to indicate how or if this detail would be included in the data extraction sheet. I believe this is a very important detail to include, "return to work" and "return to pre-injury work" are not the same outcome, I believe that there needs to be some information somewhere within the protocol that recognizes that difference, and describes how you intend to address/capture that data during data extraction. Additionally, in multiple sections this protocol still appears to be using "brain injury" and "Traumatic Brain Injury" interchangeably, which is extremely confusing to the reader. These two populations are not interchangeable. While I noticed that the title was clarified, there are still multiple instances throughout the manuscript which have not been addressed. For example, brain injury is included twice in the introduction: Line 19-20, "…is a significant challenge for brain injury survivors" Line 23, "context of RTW after brain injury" Both of these statements are accurate, but why not specify "traumatic brain injury" instead of "brain injury", which would also include individuals with acquired brain injury (e.g., stroke), individuals whom you have excluded from this review? Traumatic brain injury is not mentioned until the middle of the methods section. If this is a manuscript about WA's among survivors of traumatic brain injury, then the author needs to be sure to use "traumatic brain injury" instead of "brain injury". For example, "Traumatic Brain Injury" should be a keyword instead of "Brain Injury", the second strength of this review should say "…engaging traumatic brain injury survivors…", etc. Finally, I firmly believe that the peer-review process should be about content and not grammar, however the frequency with which grammar and punctuation errors occur in this manuscript is very difficult to ignore. I recommend that the authors consider sharing

	this manuscript with someone who can assist with these issues. There are many sentences that don't appear to be fully formed or finished, and quite a lot of inconsistency with punctuation. I have provided several examples below, but not a comprehensive list. Page 3, line 52, "rehabilitation professional" should be plural to be consistent with the other terms P3, L55: "engaging brain injury survivors"; does this mean your research will engage ALL brain injury survivors (including those with acquired brain injury) or just those with TBI? Please consider making references to "brain-injury" survivors specific to whatever population you will ultimately include in your study. P3 L63-65: are you trying to say that there is a limited number of high-level evidence within the vocab rehabilitation literature with respect to ALL environmental variables, or just for variations in disability provision? Please clarify. Implications P3 L68-70: Both of these sentences are way too broad. These implications could apply to several studies, as you have not specified how they would provide information specific to the TBI population. As an example in your second implication, you say "the most effective workforce accommodations can provide employment opportunities to people with disabilities and vulnerabilities"…so are you saying that the research done in your review will report the "most effective" accommodations for ALL individuals with disabilities and vulnerabilities, or just those within the TBI population? I would disagree with the final implication as currently written. P4 L92: These are examples, so e.g. should be used, not i.e. P4 L107: Again, these are examples, need to use e.g. and not i.e. P5 L132: I had to Google the definition of pink-collar employee because I had never heard of it. Gaudette, 1998 describes "pink-collar" occupations as female-dominated occupations, including clerical, sales, and service professions. As people outside of Canada may not be familiar with the term "pink-collar", I would suggest that you revise this sentence for clarification. For example: "…while individuals in female-dominated "pink-collar" positions are significantly more likely….." or "…while pink-collar employees (female-dominated occupations such as clerical, sales, and service professions". P5 L38: If you state that there are "ample reviews" that have investigated the role of work-related interventions on RTW…..then you need to cite at least one of those reviews at the conclusion of the sentence. P5 L156: "the long-term sustenance of this return". I believe that you mean the long-term sustainability…and if that is the case, I would just say, "and long-term sustainability after RTW". According to the dictionary, sustenance is "food and drink regarded as a source of strength; nourishment."….so I would recommend rephrasing the four areas where you used the phrase "sustenance of this return" in the manuscript.

	P5 L163: You should not assume that the reader knows what PROSPERO is. "….was registered with PROSPERO, an international database of prospectively registered systematic reviews in health and social care (citation for PRISMA website)" P6 L170: Also should not assume that the reader knows what PICO is….Population, Intervention, Comparison, Outcome framework. P7 L213: I'm wondering why interventions that include WAs (of any kind) will be included in the second level screen. If the point of this review is to assess the effectiveness of workplace accommodations, then wouldn't you want to ensure to screen abstract/titles that are specific to workplace accommodations? This seems like you would catch any article with an RTW outcome, and then exclude quite full-text articles due to not having an intervention…this does not seem very efficient. P7 L215: Should this be an operational definition of 'accommodation' or 'workplace accommodation'? P7 L228: Here you say you will stratify by "sex", throughout the introduction you were using the term "sex/gender". Personally, I think you should pick either sex or gender (rather than say "sex/gender"), but whatever you choose to do, please make sure you are consistent throughout the document. P7 L226-228: Here I would say you might want to also specify that you are also including all types of injury mechanisms (blunt vs. penetrating TBI). There are many studies that exclude patients with penetrating TBI due to their different outcomes, I would make sure you specify that you intend to include both. P7 L251: I don't believe that "inputted" is the right verb tense here. P9 L316: This statement in inaccurate as written: "Given that brain injury occurs frequently among working age individuals,"…not all brain injuries occur frequently among working age individuals (In the US, the majority of strokes occur in individuals 65 or older), traumatic brain injuries do. I would suggest revising this sentence to be consistent with your sentence on page 4, lines 80-81. P9 L17: "In countries where reasonable workplace accommodations are legislated"…what is a "reasonable" workplace accommodation? Would suggest omitting the word "reasonable" P9 L346-352: I'll echo my previous comment: your review is only looking at individuals between the ages of 18-65, so why are you making any statements about the aging population? You excluded the aging population from your study: whether workplaces need to become more accommodating of older workers and the statistics you provide about how old Canada's population is going to be in 2031 is not relevant to your study. I appreciate that you deleted the final sentence of this paragraph after my previous comment, but you left an entire paragraph still devoted to discussing workforce issues in a population you excluded from your study. I would suggest you omit these lines.

REVIEWER	Daniele Evelin Alves

	Work Research Institute, Oslo Metropolitan University, Norway
REVIEW RETURNED	09-Nov-2020

GENERAL COMMENTS	Review report: Assessing the effectiveness of workplace accommodations in facilitating return-to-work after brain injury – a systematic review protocol. Recommendation: Major revision Reviewer's summary of the article: Comments provided during last review (still relevant): The protocol is well-written and addresses an important research gap in the literature: The effectiveness of different specific work accommodations on return-to-work following acquired brain injury. However interesting and important, I have some important concerns related to the operationalization of terms in a multi-disciplined research field, as well as the ambitious scope of the future review. Firstly, the role of work accommodation in facilitation return-to-work among employees with acquired brain injury has been studied in various disciplines such as medicine, psychology, social work, rehabilitation, etc… Disciplines use different words to study work accommodation and return-to-work. It would be helpful to readers if the authors could provide a more extensive list of subject headings and keywords, and explain how exactly they were combined (for instance: how was the PICO framework modified, and how were words combined? Did you include "work retention" OR "return-to-work" OR "work tenure")? This could be done by including an appendix, or explaining the search words and strategy more clearly on line 161. As one of the authors of a recent systematic review on work and workplace characteristics related to work retention/return-to-work among employees with brain injuries, I was initially surprised by the wide variation of terms necessary to capture our research question across disciplines when conducting our data search. Our review (Alves et al., 2020) was not cited in the present protocol, and I strongly recommend the authors to read it as well as its reference list when revising the protocol. My intention is not to misuse my role as a reviewer in order to be cited, but to make sure that the protocol is up-to-date and complete. As far as I can see, Alves et al (2000, se full reference below) includes updated research that should be included in the protocol, as it includes work accommodations and brain injury (even though its scope is wider in terms of exposure and target group). The protocol describes ambitious goals (lines 147-151). Synthesizing both quantitative and qualitative evidence in the field is no easy task. Please consider using the software Covidence. Full reference of recent review: Alves DE, Nilsen W, Fure SCR, et al. What characterises work and workplaces that retain their employees following acquired brain injury? Systematic review. Occup Environ Med 2020;77(2):122-30. New comments, November 2020: Search strategy: Please include your patient population (TBI) among the search words in your search strategy, in accordance with the search words in the Appendix. Otherwise your search strategy would appear too broad. Comments provided in my last review report (this time it concerns another study): "The background section could be more precise in terms of clarifying literature on work accommodations including groups other than employees with TBI" (e.g., lines 132-134: "pink-

	collar employees: this is only one study (so “It has been shown” may confuse readers to believe that this finding has been found in several studies- consider in stead “Frances and colleagues found that …. pink-collar workers with musculoskeletal disease”. I still do not understand how you will synthesize findings from high-quality and low-quality quantitative studies (e.g., will findings from a high-quality study weigh more than those from a low-quality study?). Likewise, in terms of high- and low-quality qualitative studies. Can you please expand on this point in “Data analysis, synthesis and reporting”? This brings me to a more serious concern: work accommodation is by definition unique for each employer, employee, impairment, work at hand, society. In addition you include not only paid employment, but also volunteer work. This means that you will include studies in which employees who have been in competitive employment receive work accommodation to perform sheltered work. In my understanding, this will include a vast variety of studies, some could be case studies with one person, others could involve a corporate intervention to accommodate for workers with TBI (all of which will be individually tailored). How could all these individual accomodations be synthesized in a way that advances our knowledge on the work accomodations for employees with TBI, and can be transferred to new practical recommendations for this patient group and their employers? In addition, you wish to include sub-group analyses on gender. This is all very ambitious, and I cannot understand how you will manage to do more than map the studies on the subject, as is currently described in the protocol. Perhaps mapping the subject is enough in terms of the aim of the review given the complexity of the methods included, patient severity, heterogeneity of workplaces as well as work accomodations included in the studies? I suggest that you either explain more about how you will weigh or simply present studies (high and low-quality) within qualitative and quantitative categories in a convincing way, or consider a systematic mapping review, instead of a systematic review. “All in all, I believe that this protocol and its future review may become an important contribution to our research community”. For now, I am not convinced that it will advance our understanding of work accomodations among employees with TBI. Perhaps, the authors could decrease the scope of the review (for instance, by only mapping the studies available on work accomodations for TBI employees) or explain how a coherent synthesis of so much unique data, studies and methods will be done. Good luck with this important work.

VERSION 2 – AUTHOR RESPONSE

Reviewer #2 - Mirinda Gormley

1. I double checked several of my first comments against the marked version of the draft and was surprised how few changes were incorporated into the document. I asked several questions about the data extraction portion, specifically adding more detail to whether country of origin would be captured, and years of data collection. You responded that the variables would be included in the extraction sheet, yet without access to that extraction sheet my question remains. It would be helpful if the data extraction section included these details. For example, what will be captured for study setting? Is this going to be a description of the population, the years the study was conducted, the university

conducting the study? I would suggest revising this to include those details "…study setting (e.g., source of population, years for data analyzed, country of study, etc.)

Response: We have now included these details, previously outlined in our cover letter, within the manuscript itself. The changes have been made on page 8, Lines 278-281.

2. Similarly, I asked the how you would distinguish between "return to work" and "return to pre-injury work", as I believe that would provide important context for your discussion. The response was that the data extraction sheet was designed to be sensitive to this information, however there were no updates made to indicate how or if this detail would be included in the data extraction sheet. I believe this is a very important detail to include, "return to work" and "return to pre-injury work" are not the same outcome, I believe that there needs to be some information somewhere within the protocol that recognizes that difference, and describes how you intend to address/capture that data during data extraction.

Response: We have now included these details, previously outlined in our cover letter, within the manuscript itself. The changes have been made on page 8, Lines 287-292

3. Additionally, in multiple sections this protocol still appears to be using "brain injury" and "Traumatic Brain Injury" interchangeably, which is extremely confusing to the reader. These two populations are not interchangeable. While I noticed that the title was clarified, there are still multiple instances throughout the manuscript which have not been addressed. For example, brain injury is included twice in the introduction.
a. Line 19-20, "…is a significant challenge for brain injury survivors"
b. Line 23, "context of RTW after brain injury"
Both of these statements are accurate, but why not specify "traumatic brain injury" instead of "brain injury", which would also include individuals with acquired brain injury (e.g., stroke), individuals whom you have excluded from this review? Traumatic brain injury is not mentioned until the middle of the methods section. If this is a manuscript about WA's among survivors of traumatic brain injury, then the author needs to be sure to use "traumatic brain injury" instead of "brain injury". For example, "Traumatic Brain Injury" should be a keyword instead of "Brain Injury", the second strength of this review should say "…engaging traumatic brain injury survivors…", etc.

Response: We have now specifically used the word "traumatic" in front of "brain injury" or "TBI" throughout the manuscript.
4. Finally, I firmly believe that the peer-review process should be about content and not grammar, however the frequency with which grammar and punctuation errors occur in this manuscript is very difficult to ignore. I recommend that the authors consider sharing this manuscript with someone who can assist with these issues. There are many sentences that don't appear to be fully formed or finished, and quite a lot of inconsistency with punctuation. I have provided several examples below, but not a comprehensive list.

a. Page 3, line 52, "rehabilitation professional" should be plural to be consistent with the other terms

Response: We have made a sincere effort to re-proof the manuscript.
This specific revision has been made (Page 3, Line 56).
b. P3, L55: "engaging brain injury survivors"; does this mean your research will engage ALL brain injury survivors (including those with acquired brain injury) or just those with TBI? Please consider making references to "brain-injury" survivors specific to whatever population you will ultimately include in your study.

Response: We have made this revision (Page 3, Lines 59-60).

c. P3 L63-65: are you trying to say that there is a limited number of high-level evidence within the vocab rehabilitation literature with respect to ALL environmental variables, or just for variations in disability provision? Please clarify.

Response: Additional clarification for the 3rd point under limitations has been added. The point we are make here is - We recognize that, based on our current understanding of the literature, support for specific workplace accommodations may be limited across studies ranked as high level of evidence (Page 3, Lines 70-71).

d. Implications P3 L68-70: Both of these sentences are way too broad. These implications could apply to several studies, as you have not specified how they would provide information specific to the TBI population. As an example in your second implication, you say "the most effective workforce accommodations can provide employment opportunities to people with disabilities and vulnerabilities"…so are you saying that the research done in your review will report the "most effective" accommodations for ALL individuals with disabilities and vulnerabilities, or just those within the TBI population? I would disagree with the final implication as currently written.

Response: Based on editorial requirements, this section has now been removed.

e. P4 L92: These are examples, so e.g. should be used, not i.e

Response: We have made this revision (Page 4, Line 99).

f. P4 L107: Again, these are examples, need to use e.g. and not i.e.

Response: We have made this revision (Page 4, Line 114).

g. P5 L132: I had to Google the definition of pink-collar employee because I had never heard of it. Gaudette, 1998 describes "pink-collar" occupations as female-dominated occupations, including clerical, sales, and service professions. As people outside of Canada may not be familiar with the term "pink-collar", I would suggest that you revise this sentence for clarification. For example: "…while individuals in female-dominated "pink-collar" positions are significantly more likely….." or "…while pink-collar employees (female-dominated occupations such as clerical, sales, and service professions".

Response: We have made these revisions (Page 5, Lines 143-149).

h. P5 L38: If you state that there are "ample reviews" that have investigated the role of work-related interventions on RTW…..then you need to cite at least one of those reviews at the conclusion of the sentence.

Response: We have added citations to support the narrative (Page 5, Line 157).

i. P5 L156: "the long-term sustenance of this return". I believe that you mean the long-term sustainability…and if that is the case, I would just say, "and long-term sustainability after RTW". According to the dictionary, sustenance is "food and drink regarded as a source of strength; nourishment."….so I would recommend rephrasing the four areas where you used the phrase "sustenance of this return" in the manuscript.

Response: We have made these revisions as requested (Page 3, Line 54 & Page 6, Line 177).

j. P5 L163: You should not assume that the reader knows what PROSPERO is. "….was registered with PROSPERO, an international database of prospectively registered systematic reviews in health and social care (citation for PRISMA website)"

Response: We have partly made this revision. It is unclear why we are asked to cite the PRISMA website when discussing PROSPERO? (Page 6, Lines 187-189).

k. P6 L170: Also should not assume that the reader knows what PICO is….Population, Intervention, Comparison, Outcome framework.

Response: We have made this revision as requested (Page 6, Lines 196-199).

l. P7 L213: I'm wondering why interventions that include WAs (of any kind) will be included in the second level screen. If the point of this review is to assess the effectiveness of workplace accommodations, then wouldn't you want to ensure to screen abstract/titles that are specific to workplace accommodations? This seems like you would catch any article with an RTW outcome, and then exclude quite full-text articles due to not having an intervention…this does not seem very efficient.

Response: Based on our familiarity with the return-to-work literature specific to TBI, workplace accommodations are generally not mentioned in abstracts. Including 'workplace accommodations' in level I screen we felt would result in the erroneous elimination of articles that could have met the inclusion criteria (Page 7, Lines 254-258).

m. P7 L215: Should this be an operational definition of 'accommodation' or 'workplace accommodation'?

Response: We have made this revision as requested (Page 7, Line 259).

n. P7 L228: Here you say you will stratify by "sex", throughout the introduction you were using the term "sex/gender". Personally, I think you should pick either sex or gender (rather than say "sex/gender"), but whatever you choose to do, please make sure you are consistent throughout the document.

Response: We have chosen to retain both sex and gender (if possible with the data extracted) for the purposes of the review. This is difficult because biological sex and gender terms are used interchangeably in the literature. Some authors indicate they are looking at sex but then list responses that reflect gender and vice versa. We will likely be commenting on this in the manuscript. How sex and gender are used in the studies and to what extent sex and gender are given explicit consideration given the gendered nature of work may be an original contribution to this literature.

o. P7 L226-228: Here I would say you might want to also specify that you are also including all types of injury mechanisms (blunt vs. penetrating TBI). There are many studies that exclude patients with penetrating TBI due to their different outcomes, I would make sure you specify that you intend to include both.

Response: We have made this revision as requested (Page 8, Lines 271-272).

p. P7 L251: I don't believe that "inputted" is the right verb tense here.

Response: We have provided an alternative word and completed this revision as requested (Page 8, Line 299).

q. P9 L316: This statement in inaccurate as written: "Given that brain injury occurs frequently among working age individuals,"…not all brain injuries occur frequently among working age individuals (In the US, the majority of strokes occur in individuals 65 or older), traumatic brain injuries do. I would suggest revising this sentence to be consistent with your sentence on page 4, lines 80-81.

Response: We have revised this statement as requested (Page 10, Line 384-385).

r. P9 L17: “In countries where reasonable workplace accommodations are legislated”…what is a “reasonable” workplace accommodation? Would suggest omitting the word “reasonable”

Response: We have revised this statement as requested (Page 10, Line 403).

s. P9 L346-352: I’ll echo my previous comment: your review is only looking at individuals between the ages of 18-65, so why are you making any statements about the aging population? You excluded the aging population from your study: whether workplaces need to become more accommodating of older workers and the statistics you provide about how old Canada’s population is going to be in 2031 is not relevant to your study. I appreciate that you deleted the final sentence of this paragraph after my previous comment, but you left an entire paragraph still devoted to discussing workforce issues in a population you excluded from your study. I would suggest you omit these lines.

Response: We have omitted the corresponding narrative on page 11.

Lastly, we thank you again for the thorough review and for considering our manuscript for publication. We look forward to your response.

Reviewer #3 - Daniele Evelin Alves

1. The role of work accommodation in facilitation return-to-work among employees with acquired brain injury has been studied in various disciplines such as medicine, psychology, social work, rehabilitation, etc… Disciplines use different words to study work accommodation and return-to-work. It would be helpful to readers if the authors could provide a more extensive list of subject headings and keywords, and explain how exactly they were combined (for instance: how was the PICO framework modified, and how were words combined? Did you include “work retention” OR “return-to-work” OR “work tenure”)? This could be done by including an appendix, or explaining the search words and strategy more clearly on line 161.

Response: We have now expanded on this section by outlining how various operators were used to combine search words in order to achieve the balance of sensitivity with specificity required. (Page 6/7, Lines 196-218).

2. As one of the authors of a recent systematic review on work and workplace characteristics related to work retention/return-to-work among employees with brain injuries, I was initially surprised by the wide variation of terms necessary to capture our research question across disciplines when conducting our data search. Our review (Alves et al., 2020) was not cited in the present protocol, and I strongly recommend the authors to read it as well as its reference list when revising the protocol. My intention is not to misuse my role as a reviewer in order to be cited, but to make sure that the protocol is up-to-date and complete. As far as I can see, Alves et al (2000, se full reference below) includes updated research that should be included in the protocol, as it includes work accommodations and brain injury (even though its scope is wider in terms of exposure and target group).

Response: We have cited the review (Page 5, Lines 164-169).

3. The protocol describes ambitious goals (lines 147-151). Synthesizing both quantitative and qualitative evidence in the field is no easy task. Please consider using the software Covidence.

Response: We are currently exploring Covidence to progress the review.

4. Search strategy: Please include your patient population (TBI) among the search words in your search strategy, in accordance with the search words in the Appendix. Otherwise your search strategy would appear too broad.

Response: We have now added an overview within the narrative to provide a general sense of the search terms used to define the patient populations (TBI); changes are reflected on Page 6, Lines 202-205. The list of search terms, included with the manuscript, of course provides a more comprehensive list.

5. Comments provided in my last review report (this time it concerns another study): “The background section could be more precise in terms of clarifying literature on work accommodations including groups other than employees with TBI” (e.g., lines 132-134: “pink-collar employees: this is only one study (so “It has been shown” may confuse readers to believe that this finding has been found in several studies- consider in stead “Frances and colleagues found that …. pink-collar workers with musculoskeletal disease”.

Response: We have made the requested revision (Page 5, Lines 143-149).

6. I still do not understand how you will synthesize findings from high-quality and low-quality quantitative studies (e.g., will findings from a high-quality study weigh more than those from a low-quality study?). Likewise, in terms of high- and low-quality qualitative studies. Can you please expand on this point in “Data analysis, synthesis and reporting”?

Response: We anticipate that there will be few controlled studies that would be rated as high quality using typical level of evidence paradigms. Given the customizable and individualized nature of workplace accommodations, we believe that the richness of the data will actually come from small cohort, case control, case-series and qualitative studies. We have outlined the analysis plan on page 9, lines 320-324 to elaborate our approach on how we intend to incorporate findings from all levels of evidence.

7. This brings me to a more serious concern: work accommodation is by definition unique for each employer, employee, impairment, work at hand, society. In addition you include not only paid employment, but also volunteer work. This means that you will include studies in which employees who have been in competitive employment receive work accommodation to perform sheltered work. In my understanding, this will include a vast variety of studies, some could be case studies with one person, others could involve a corporate intervention to accommodate for workers with TBI (all of which will be individually tailored). How could all these individual accommodations be synthesized in a way that advances our knowledge on the work accommodations for employees with TBI, and can be transferred to new practical recommendations for this patient group and their employers? In addition, you wish to include sub-group analyses on gender. This is all very ambitious, and I cannot understand how you will manage to do more than map the studies on the subject, as is currently described in the protocol. Perhaps mapping the subject is enough in terms of the aim of the review given the complexity of the methods included, patient severity, heterogeneity of workplaces as well as work accommodations included in the studies? I suggest that you either explain more about how you will weigh or simply present studies (high and low-quality) within qualitative and quantitative categories in a convincing way, or consider a systematic mapping review, instead of a systematic review.

Response: We agree in that that consolidation of both qualitative and quantitative studies in a single review will not be feasible. At the present time, we anticipate the consolidation and synthesis from qualitative and quantitative studies will be published likely as two separate reviews. Our intent is to provide an overview of the state of knowledge to inform future research directions and to provide

categorical examples for the kinds of accommodations have been implemented (where data is available) ad reported to be successful with respect to RTW outcomes. Please see out changes on page 9, lines 325-338.

VERSION 3 – REVIEW

REVIEWER	Alves, Daniele Oslo Metropolitan University
REVIEW RETURNED	02-Mar-2021

GENERAL COMMENTS	The authors have addressed all my major concerns in this version of the protocol. I believe that this ambitious review will contribute to the field, and I look forward to reading it. Please consider a small change in the abstract's objectives. In the paper, there are three objectives. In the abstract, only two of them are stated, while the third objective (gender differences) is not mentioned. I suggest changing the text in the abstract from "The primary objective" to "Objectives of this protocol are…". This way, the authors can mention only two of the objectives in the abstract, if they wish, or even include the third objective. Consider also adding "the heterogeneity of the methods included in the review" in limitations on page 3, line 61.